

Sets and Venn Diagrams Using Real-World Data

Instructor Lesson Plan

Math 140 – Math for Elementary Teachers

Lesson Information

Course: Math 140 – Math for Elementary Teachers
Topic: Applications of Sets and Venn Diagrams Using Real-World Data
Class Time: Approximately 2 hours
Setting: Computer lab with Microsoft Excel
Materials: Student activity handout, simplified Excel dataset, solutions handout

Prior Preparation: Basic set notation, unions, intersections, complements, Venn diagrams, and basic Excel filtering.

Learning Objectives

Students will be able to:

- organize real-world data into sets and use Excel filters to identify and count observations;
- represent data using two-set and three-set Venn diagrams;
- interpret unions, intersections, and complements using data;
- solve a basic Venn diagram problem without Excel.

Lesson Plan

Time	Activity	Plan
0–10 min	Review	Review set notation, Venn diagrams, and the purpose of the real-world dataset.
10–25 min	Part 1: By Hand	Organize the 10-observation sample into sets and complete a two-set Venn diagram.
25–35 min	Part 2: Excel	Open the full dataset and briefly review the columns and Excel filters.
35–60 min	Part 3: Two Sets	Use Excel to find the required counts and construct the two-set Venn diagram by hand.
60–75 min	Interpretation	Use the completed diagram to interpret intersections, unions, and complements.
75–100 min	Part 4: Three Sets	Use Excel to investigate three sets and construct a three-set Venn diagram, beginning with the center intersection.
100–110 min	Part 5: No Excel	Independently complete the final Venn diagram problem without Excel.
110–120 min	Reflection	Complete the final reflection and connect sets and Venn diagrams to organizing real-world data.

Assessment

Student understanding will be assessed through **Excel filtering and data counts, completed Venn diagrams, interpretation of set notation, an independent Venn diagram application, and the final reflection.**

Lesson Emphasis: Excel helps students organize and examine a larger dataset, while constructing and interpreting the Venn diagrams by hand reinforces the mathematics.